Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/12/2024
PI Name:	Zea, Luis Ph.D.		
Project Title:	Characterization of Fungal Biofilm (Molds) Formation, Growth, and Gene Expression on Different Materials in Microgravity		
Division Name:	Physical Sciences, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80309-0429	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	08/23/2021	**End Date:**	04/30/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Gorti, Sridhar	**Contact Phone:**	256 544-0158
Contact Email:	sridhar.gorti@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 04/30/2024 per NSSC information (Ed., 10/31/23). NOTE: End date changed to 10/31/23 per NSSC information (Ed., 8/21/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Klaus, David Ph.D. (University of Colorado, Boulder) Moeller, Ralf Ph.D. (Deutsches Zentrum Fuer Luft- Und Raumfahrt E.V., Germany) Muecklich, Frank Ph.D. (Saarland University, Germany) Stodieck, Louis Ph.D. (University of Colorado, Boulder) Flores, Pamela B.S. (University of Colorado Boulder)		
Grant/Contract No.:	80NSSC21K1950		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Project related to"Characterization of Biofilm Formation, Growth, and Gene Expression on Different Materials and Environmental Conditions in Microgravity" (grant numbers 80NSSC17K0036; NNX17AC21G), with same Principal Investigator, Dr. Luis Zea. Biofilm formation and its concomitant risk of causing equipment malfunction and human illnesses is a problem that needs to be addressed to enable safe long-term human space exploration. With that objective, this proposal aims to characterize biofilm growth during one International Space Station (ISS) increment in terms of mass, thickness, morphology, and associated gene expression using different spaceflight relevant material substrata. The experiment is designed to assess the expression of genes associated with conferring microorganisms with resistance to oxidative stress, acidity, and antimicrobials in biofilms formed in space, and to determine physical mechanisms of material/microorganisms interaction in biofilms. The experiment will be conducted using spaceflight-compatible well plates housing substrata coupons exposed to different microorganisms and media. Post-flight analyses of fixed samples include confocal laser scanning microscopy and RNA sequencing. The significance of this project stems from gaining valuable new knowledge that may lead to improved methods and/or material properties for controlling biofilm formation during long duration space missions. The selection of fungal species and substrata material make the results applicable to a) the safety and maintenance of crewed spacecraft, b) planetary protection, c) mitigation of biofilm- associated illnesses on the crew, and d) biofilm-associated illnesses on Earth, thus making it relevant not only to this Research Announcement, but also to NASA's Space Biology Program, and Human Research Roadmap.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Biofilms are associated with multiple illnesses, including endocarditis (bacterial infection of cardiac tissue), cystitis (a urinary tract infection), and otitis media (an inflammatory disease of the middle ear), as well as several other hospital-acquired infections. Understanding how these biofilms form and their relationship with resistance to antibiotics can help us better prepare novel solutions to these problems on Earth.
Task Progress:	This was one of two initial NASA Biological and Physical Sciences (BPS) dual Biology-Physical Sciences supported payloads. The Space Biofilms reflight project has been successfully completed and under budget. Microgravity did not have a strong effect on biofilm shape nor morphology of P. rubens. The nanograss material was able to significantly reduce fungal biofilm formation both on Earth and microgravity. Eight undergraduate and seven graduate students were involved in the project. This project resulted in one book, one Ph.D. thesis, one Master's thesis, four undergraduate theses, 7 peer-reviewed papers, and 31 conference presentations, abstracts, and posters. In addition to uploading our data to NASA's GeneLab and Physical Sciences Informatics (PSI), we voluntarily sent our biological samples to NASA's Space Biology Biospecimen Sharing Program so other scientists can perform research on our flown samples, and double-check our work.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Articles in Peer-reviewed Journals	Flores P, Luo J, Mueller DW, Muecklich F, Zea L. "Space biofilms – An overview of the morphology of Pseudomonas aeruginosa biofilms grown on silicone and cellulose membranes on board the international space station." Biofilm. 2024 Feb 5:100182. https://doi.org/10.1016/j.bioflm.2024.100182 ; PMID: 38370151; PMCID: PMC10869243 , Feb-2024
Articles in Peer-reviewed Journals	Herrera-Jordan K, Pennington P, Zea L. "Reduced Pseudomonas aeruginosa cell size observed on planktonic cultures grown in the International Space Station." Microorganisms. 2024 Feb 16;12(2):393. https://doi.org/10.3390/microorganisms12020393 ; PMID: 38399797; PMCID: PMC10892763 , Feb-2024
Articles in Peer-reviewed Journals	Mason CE, Green J, Adamopoulos KI, Afshin EE, Baechle JJ, Basner M, Bailey SM, Bielski L, Borg J, Borg J, Broddrick JT, Burke M, Caicedo A, Castañeda V, Chatterjee S, Chin C, Church G, Costes SV, De Vlaminck I, Desai RI, Dhir R, Diaz JE, Etlin SM, Feinstein Z, Furman D, Garcia-Medina JS, Garrett-Bakelman F, Giacomello S, Gupta A, Hassanin A, Houerbi N, Irby I, Javorsky E, Jirak P, Jones CW, Kamal KY, Kangas BD, Karouia F, Kim J, Kim JH, Kleinman A, Lam T, Lawler JM, Lee JA, Limoli CL, Lucaci A, MacKay M, McDonald JT, Melnick AM, Meydan C, Mieczkowski J, Muratani M, Najjar D, Othman MA, Overbey EG, Paar V, Park J, Paul AM, Perdyan A, Proszynski J, Reynolds RJ, Ronca AE, Rubins K, Ryon KA, Sanders LM, Glowe PS, Shevde Y, Schmidt MA, Scott RT, Shirah B, Sienkiewicz K, Sierra M, Siew K, Theriot CA, Tierney BT, Venkateswaran K, Hirschberg JW, Walsh SB, Walter C, Winer DA, Yu M, Zea L, Mateus J, Beheshti A. "A second space age spanning omics, platforms and medicine across orbits." Nature. 2024 Aug;632(8027):995-1008. Review. https://doi.org/10.1038/s41586-024-07586-8 ; PubMed PMID: 38862027; PubMed Central PMCID: PMC12366838 , Aug-2024